

AI Techniques for Wheelchair-Accessible Urban Navigation

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Abstract

Code —

<https://github.com/francescagrasso02/TechniquesOfAI>

Introduction

Currently, route planning applications are mainly based on static criteria such as minimum distance or minimum estimated walking time (Darko, Hossain, and Rahman 2022). However, for wheelchair users, these parameters are not sufficient to be able to find a route that they can take independently. Factors such as narrow sidewalks, steep ramps or inaccessible kerbs may make the route impossible for people with mobility impairments (Darko, Hossain, and Rahman 2022). Recent studies on urban mobility in Brussels highlight problems related to inaccessible kerbs, discontinuities in the tactile pavement and obstacles present on sidewalks (Dobruszkes, Grandjean, and Nihoul 2025).

Problem Formulation

This work proposes a route planning system that focuses on accessibility for wheelchair users in the city of Brussels. Using real OpenStreetMap data, this system models the urban pedestrian network as a graph, taking into account crucial accessibility factors such as slope, sidewalk width, surface type and kerb accessibility (Kaselouris et al. 2025).

Three distinct artificial intelligence approaches are implemented to solve this problem: heuristic search algorithms, such as Dijkstra and A*, Constraint Satisfaction Problems (CSP) and Markov Decision Processes (MDP). These three approaches, which address the same problem from different perspectives, will be compared experimentally and qualitatively to analyze their trade-offs, strengths, and limitations.

Formal Representation

The pedestrian network is modeled as a weighted graph:

$$G = (V, E)$$

where:

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- V represents navigable pedestrian nodes extracted from OpenStreetMap
- E represents sidewalk or street segments connecting adjacent nodes

Each edge $e \in E$ contains accessibility-related attributes:

$$e = (\text{length}, \text{slope}, \text{width}, \text{kerb}, \text{wheelchair}, \text{surface})$$

including:

- **length**: street length in metres
- **slope**: gradient percentage
- **width**: sidewalk width in metres
- **kerb**: kerb accessibility type
- **wheelchair**: wheelchair accessibility tag (*yes, no, limited, unknown*)
- **surface**: surface type

The objective is to compute an accessible path:

$$P = (v_1, v_2, \dots, v_n)$$

between an origin node v_1 and a destination node v_n , while minimizing traversal cost, which is primarily based on path length:

$$\min \sum_{e \in P} c(e)$$

The routing process is subject to several accessibility constraints related to pedestrian mobility, including sidewalk slope, minimum width requirements, kerb accessibility, and wheelchair accessibility tags. The system applies strict constraints by default, but switches to relaxed constraints when no feasible accessible route can be found.

Suitability for Multiple AI Techniques

This problem allows the application of search algorithms such as Dijkstra and A*, since the pedestrian network can be represented as a weighted graph for finding minimum-cost accessible routes. Constraint Satisfaction Problems can also be applied to ensure that the route satisfies specific accessibility constraints. Finally, Markov Decision Processes can be used to model uncertainty in accessibility conditions due to the presence of incomplete accessibility information in OpenStreetMap data.

Methodological Approaches

Approach A: Search Algorithms

Approach B: Constraint Satisfaction Problems

Approach C: Markov Decision Processes

Experimental Evaluation

Qualitative Comparison

Conclusion

References

Darko, A.; Hossain, M. S.; and Rahman, M. 2022. Reinforcement Learning-Based Outdoor Path Planning for Wheelchair Users. *Sensors*, 22(19): 7429.

Dobruszkes, F.; Grandjean, M.; and Nihoul, A. 2025. Towards Public Transport for All in Brussels: Testing Inclusive Design-Based Scenarios. *Brussels Studies*, (206).

Kaselouris, E.; et al. 2025. Accessibility Routing Model for the Safe Navigation of Vulnerable Road Users in Urban Environments. *Sustainability*, 17(2).